



Connect Shiksha Coding Classes

# Web Development Curriculum

HTML5, CSS3, JavaScript, React JS, and Backend roadmap for classroom teaching, student sharing, revision, and project practice.

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Student-ready PDF copy with matching Connect Shiksha branding.

| Modules | Lessons | Projects / Practicals | Audience              |
|---------|---------|-----------------------|-----------------------|
| 6       | 141     | 139                   | Teachers and students |

# 1. HTML5

Lessons: 35 | Capstone / main project: Netflix Website

| No. | Topic                | Details / Concepts | Project / Practical      |
|-----|----------------------|--------------------|--------------------------|
| 1   | Introduction         | Introduction       | Practice task            |
| 2   | Editors(VS CODE)     | Editor             | Hello World Program      |
| 3   | Basic Structure      | Basic              | Hello World Program      |
| 4   | Doctype              | Doctype            | Basic Resume Info        |
| 5   | Elements             | Elements           | Basic Resume Add details |
| 6   | Attributes           | Attributes         | Basic Resume Image       |
| 7   | Headings             | Headings           | Essay Writing Using HTML |
| 8   | Paragraph            | Paragraph          | Essay Writing Using HTML |
| 9   | Text Formatting      | Formating          | Apply All Formatting     |
| 10  | Comments             | Comments           | Section Comments         |
| 11  | Semantics            | semantic           | 5 Full Form and One Song |
| 12  | CSS Style            | Style              | Link CSS To HTML         |
| 13  | Links                | Links              | IG, FB,Google Link       |
| 14  | Image                | Image              | Adding Image in HTML     |
| 15  | Favicon              | Favicon            | Adding Favicon           |
| 16  | Phone and Email Link | Email              | Link Phone And Email     |
| 17  | Marquee              | Marquee            | Football Animation       |
| 18  | Lists                | Lists              | List of Movies           |
| 19  | Table                | Table              | Database of Friends      |
| 20  | Block & Inline       | Block & Inline     | Layout                   |
| 21  | Iframe               | Iframe             | 4 Web Page in one Page   |

| No. | Topic           | Details / Concepts | Project / Practical      |
|-----|-----------------|--------------------|--------------------------|
| 22  | Audio           | Audio              | Audio Add in Web Page    |
| 23  | Video           | Video              | Video Add in Web Page    |
| 24  | Document        | Document           | Document Add in Web Page |
| 25  | Google Map      | Map                | Map Add in Web Page      |
| 26  | Youtube Video   | Yt Video           | Yt Video Add in Web Page |
| 27  | Class           | Class              | Style Page Using Class   |
| 28  | ID              | Id                 | Nav Bar Project          |
| 29  | Entities        | Entities           | Create Copyright Footer  |
| 30  | Symbols         | Symbols            | Link Phone And Email     |
| 31  | Emojis          | Emojis             | Create Emoji Web Page    |
| 32  | Form            | Form               | Registration Form        |
| 33  | Head            | Head               | ShortCut of Head         |
| 34  | Viewport        | Viewport           | Responsive Website       |
| 35  | Summary of Html | Summary            | Questions and doubts     |

## 2. CSS3

Lessons: 31 | Capstone / main project: Portfolio Website

| No. | Topic         | Details / Concepts | Project / Practical           |
|-----|---------------|--------------------|-------------------------------|
| 1   | Introduction  | Intro              | Practice task                 |
| 2   | Syntax        | Syntax             | Colorful Web Page             |
| 3   | Selectors     | Selectors          | 3 Box Using Differents color  |
| 4   | Comments      | Comments           | Using Comments in CSS         |
| 5   | Colors        | Colors             | Creating Rainbow Name         |
| 6   | Background    | Background         | Using Background in CSS       |
| 7   | Borders       | Borders            | Creating Status Boxes         |
| 8   | Padding       | Padding            | Google Front Page             |
| 9   | Margin        | Margin             | Google Front Page             |
| 10  | Box Model     | Box Model          | Mathometies Calculation       |
| 11  | Text          | Text               | Write ABCD in Cursive Writing |
| 12  | Fonts         | Fonts              | Using Google fonts            |
| 13  | Icons         | Icons              | Social Media Icons            |
| 14  | Table Styling | Table              | Style a Table                 |
| 15  | Box Shadow    | Box Shadow         | 3 Boxes Using Shadow          |
| 16  | Display       | Display            | Using Display Option          |
| 17  | Position      | Position           | Sticky menu and Whatsapp      |
| 18  | Gradients     | Gradients          | National Flag                 |
| 19  | Float         | Float              | 3 Section with Float Image    |
| 20  | Z-Index       | Z-index            | Book Effects                  |
| 21  | Object-fit    | Object fit         | Images Objects Fits           |

| No. | Topic             | Details / Concepts | Project / Practical        |
|-----|-------------------|--------------------|----------------------------|
| 22  | Pseudo-class      | P-class            | Use of Pseudo Class        |
| 23  | Pseudo-Elements   | P-elements         | Use of Pseudo Elements     |
| 24  | 2d Transformation | 2d Transform       | All Effects Use            |
| 25  | 3d Transformation | 3d Transform       | All Effects Use            |
| 26  | Transition        | Transition         | All Effects Use            |
| 27  | Animation         | Animation          | Creating Something         |
| 28  | Media             | Media              | Screen type styling effect |
| 29  | Flexbox           | Flexbox            | Keyboard                   |
| 30  | Grid              | Grid               | Grid Practicale            |
| 31  | CSS Summary       | Summary            | Questions And Dubtes       |

## 3. Basic Javascript

Lessons: 35 | Capstone / main project: Portfolio Website

| No. | Topic           | Details / Concepts              | Project / Practical |
|-----|-----------------|---------------------------------|---------------------|
| 1   | JS Introduction | JavaScript Introduction         | JS Web Page         |
| 2   | JS Where To     | JavaScript Where To             | JS Web Page         |
| 3   | JS Output       | JavaScript Output               | JS Web Page         |
| 4   | JS Statements   | JavaScript Statements           | JS Web Page         |
| 5   | JS Syntax       | JavaScript Syntax               | JS Web Page         |
| 6   | JS Comments     | JavaScript Comments             | JS Web Page         |
| 7   | JS Variables    | JavaScript Variables            | JS Web Page         |
| 8   | JS Let          | JavaScript Let                  | JS Web Page         |
| 9   | JS Confs        | JavaScript const                | JS Web Page         |
| 10  | JS Operators    | JavaScript Operators            | JS Web Page         |
| 11  | JS Arithmetic   | JavaScript Arithmetic           | JS Web Page         |
| 12  | JS Assignment   | JavaScript Assignment Operators | JS Web Page         |
| 13  | JS Data Types   | JavaScript Data Types           | JS Web Page         |
| 14  | JS Functions    | JavaScript Function Study Path  | JS Web Page         |
| 15  | JS Objects      | JavaScript Objects              | JS Web Page         |
| 16  | JS Events       | JavaScript Events               | JS Web Page         |
| 17  | JS Strings      | JavaScript Strings              | JS Web Page         |
| 18  | JS Arrays       | JavaScript Arrays               | JS Web Page         |
| 19  | JS Numbers      | JavaScript Numbers              | JS Web Page         |
| 20  | JS Maths        | JavaScript Math Object          | JS Web Page         |
| 21  | JS Random       | JavaScript Random               | JS Web Page         |

| No. | Topic               | Details / Concepts              | Project / Practical |
|-----|---------------------|---------------------------------|---------------------|
| 22  | JS Comparison       | JavaScript Comparison Operators | JS Web Page         |
| 23  | JS If else          | JavaScript else Statement       | JS Web Page         |
| 24  | JS Switch           | JavaScript Switch Statement     | JS Web Page         |
| 25  | JS Loop For         | JavaScript for Loop             | JS Web Page         |
| 26  | JS Loop While       | JavaScript While Loop           | JS Web Page         |
| 27  | JS Loop Do While    | JavaScript While Loop           | JS Web Page         |
| 28  | JS Typeof           | JavaScript typeof               | JS Web Page         |
| 29  | JS Type Conversion. | JavaScript Type Conversions     | JS Web Page         |
| 30  | JS Scope            | JavaScript Scope                | JS Web Page         |
| 31  | JS Hoisting         | JavaScript Scope                | JS Web Page         |
| 32  | JS Strict Mode      | JavaScript "use strict"         | JS Web Page         |
| 33  | JS this Keyword     | JavaScript this in Objects      | JS Web Page         |
| 34  | JS Classes          | JavaScript Classes              | JS Web Page         |
| 35  | JS Summary          | Class discussion                | JS Web Page         |

## 4. Advance Javascript

Lessons: 15 | Capstone / main project: Quiz Game

| No. | Topic                    | Details / Concepts                         | Project / Practical            |
|-----|--------------------------|--------------------------------------------|--------------------------------|
| 1   | JS Introduction + Output | console.log(), getElementById(), innerText | Dynamic Name Change            |
| 2   | Variables, Let & Const   | let, const, innerHTML, addEventListener()  | Theme Switcher                 |
| 3   | Operators & Data Types   | , -, *, /, %, typeof                       | Simple Calculator              |
| 4   | Functions & Events       | function, return, addEventListener()       | Button Click Counter           |
| 5   | If-Else & Comparison     | if, else, >=, <=                           | Age Verification App           |
| 6   | Switch & Loops           | switch, for, while                         | Multiplication Table Generator |
| 7   | Strings & Arrays         | length, split(), join(), push(), pop()     | Text Formatting                |
| 8   | Array Methods & Objects  | map(), filter(), push(), object            | To-Do List                     |
| 9   | Math & Random            | Math.random(), Math.floor()                | Number Predication Game        |
| 10  | DOM Manipulation         | querySelectorAll(), forEach(), display     | Image Gallery                  |
| 11  | Forms & Validation       | event.preventDefault(), value, if-else     | Storage Login Form             |
| 12  | Timers & Countdown       | setInterval(), Date()                      | Age Calculator                 |
| 13  | Search & Filtering       | filter(), input event                      | Product Search                 |
| 14  | Mini Portfolio Project   | classList.toggle()                         | Skills & Projects Toggle       |
| 15  | Final Challenge          | event listeners, arrays, if-else           | Quiz Game                      |

## 5. React JS

Lessons: 15 | Capstone / main project: Live Link Generation

| No. | Topic                     | Details / Concepts                      | Project / Practical     |
|-----|---------------------------|-----------------------------------------|-------------------------|
| 1   | React Intro & Install     | npm start, File Structure               | Hello World App         |
| 2   | JSX (HTML in JS)          | Writing HTML inside JS                  | Personal Bio Page       |
| 3   | Functional Components     | Creating & Importing Components         | Simple Button Component |
| 4   | Props (Passing Data)      | Passing data Parent to Child            | Student ID Card         |
| 5   | Events Handling           | onClick, Alert Message                  | Click Me Button         |
| 6   | useState Hook (Intro)     | State, useState Hook                    | Simple Counter (+/-)    |
| 7   | Handling Input (Forms)    | onChange, Value binding                 | Live Text Mirror        |
| 8   | Conditional Rendering     | Ternary Operator (condition ? yes : no) | Show/Hide Text Button   |
| 9   | Simple Lists (Arrays)     | map() function                          | Fruit List Display      |
| 10  | CSS in React              | className, Inline Styles                | Colorful Cards          |
| 11  | Props Drilling (Practice) | Reusing Components with different data  | Product Price Display   |
| 12  | useEffect (Very Basic)    | document.title change                   | Auto Update Title       |
| 13  | Adding Images & Icons     | Importing Images, React Icons           | Photo Gallery           |
| 14  | Simple Project Assembly   | React Routing                           | Review Previous Topics  |
| 15  | Build & Deploy            | npm run build, Netlify/Vercel Upload    | Live Link Generation    |

## 6. Backend

Lessons: 10 | Capstone / main project: Program

| No. | Topic                            | Details / Concepts | Project / Practical |
|-----|----------------------------------|--------------------|---------------------|
| 1   | React JS Set Up                  | Class discussion   | Program             |
| 2   | React JS Basics                  | Class discussion   | Program             |
| 3   | Tailwind CSS                     | Class discussion   | Program             |
| 4   | Tailwind CSS Setup With React JS | Class discussion   | Program             |
| 5   | Node JS Set Up                   | Class discussion   | Program             |
| 6   | Node JS Basic                    | Class discussion   | Program             |
| 7   | Mongo DB Set Up                  | Class discussion   | Program             |
| 8   | Mongo DB Connectivity            | Class discussion   | Program             |
| 9   | React Nodejs Mongodb Connection  | Class discussion   | Program             |
| 10  | Final Full Stack Project         | Class discussion   | Program             |